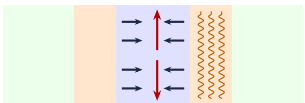
JUSTIN SUN PRIZE · JSP-000005 · EXPOSITION

How a Forced Navier–Stokes Flow Blows Up

A guided tour of OpenAI's finite-time blowup construction



Thanh Van Nguyen · Companion video to the essay (v1.0, 29 October 2026)

All mathematics is due to the authors of the source paper. This is an exposition.

The question: Navier–Stokes in three dimensions

$$\partial_t u + (u \cdot \nabla)u = \nu \Delta u - \nabla p + f, \qquad \nabla \cdot u = 0, \qquad u(\cdot, 0) = u^\circ$$

Unforced ($f \equiv 0$): global regularity

(A) on $\mathbb{R}^3$ **(B)** on the torus $\mathbb{R}^3/\mathbb{Z}^3$

smooth solutions for all time, for all smooth data

Breakdown: a force is allowed

(C) on $\mathbb{R}^3$ **(D)** on the torus

smooth data and force with *no* global smooth bounded-energy solution

Fefferman, official Clay Millennium Problem statement. Proving any one alternative resolves the problem.

The result [Paper, Theorem 1.1]

For every $\nu > 0$ there exist

a force $f \in C_c^\infty(\mathbb{R}^3 \times (0,\infty))$ and smooth u, p on $\mathbb{R}^3 \times [0,1)$ with $u(\cdot,0) = 0$, supported in a fixed compact set K, such that

$$\sup_{t<1} \|u(t)\|_{L^2} < \infty, \qquad \limsup_{t\uparrow 1} \|u(t)\|_{L^\infty} = \infty.$$

- $\Rightarrow$ alternative **(C)**; rescaling + periodizing $\Rightarrow$ **(D)**
- **Not** addressed: the unforced alternatives (A) and (B)
- Status: Lean formalization released; independent peer review pending

Why three dimensions are hard [Context]

Scaling symmetry

$$u_\lambda(x,t) = \lambda\, u(\lambda x, \lambda^2 t)$$

Energy under zoom-in:

$$\|u_\lambda\|_{L^2}^2 = \lambda^{-1}\|u\|_{L^2}^2$$

Small scales weaken the only global bound $\Rightarrow$ energy is **supercritical**.

- 1934 Leray: global weak solutions
- 1982 Caffarelli–Kohn–Nirenberg: small singular set
- 2003 Escauriaza–Seregin–Šverák: $L^\infty L^3$ regularity
- 2016 Tao: blowup for an *averaged* equation
- 2019–22 Buckmaster–Vicol, Albritton–Brué–Colombo: non-uniqueness
- 2023–26 Córdoba–Martínez-Zoroa(–Zheng): forced Euler / hypodissipative blowup

Key reformulation: choose the flow, read off the force

$$\mathcal{R}(u,p) := \partial_t u + (u\cdot\nabla)u - \nu\Delta u + \nabla p \qquad \Longrightarrow \qquad f := \mathcal{R}(u,p)$$

The real problem

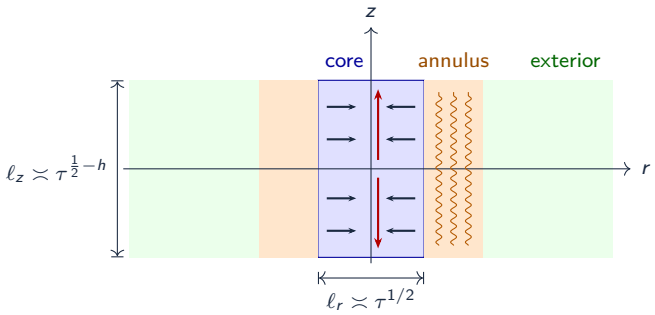
Design (u,p) that blows up at $t=1$ while $\mathcal{R}(u,p)$ **and all its derivatives** extend smoothly through $t=1$.

Flatness [Paper, eq. (3.4)]:

$$|\partial_x^\alpha \partial_t^b \mathcal{R}(u,p)| \le C_{\alpha,b,N}\, q^N \qquad \text{for every } N,$$

$q \to 0$ measures the distance to the singular point. Each term blows up; their sum must cancel to infinite order.

The collapsing vortex [Paper, §2]



- $\tau = 1 - t$; cylindrical (r, θ, z)
- core: inward spiral, axial outflow
- inflow conserves $ru_\theta \Rightarrow$ spin-up; viscosity balances it
- exterior: pure swirl solving the heat equation exactly, so zero residual
- fixed $0 < h < \frac{1}{100}$: radius shrinks faster than height

Scales of the core [Paper §2.1, §3.5; Check]

$$|u_\theta|, |u_z| \asymp \tau^{-\frac{1}{2}-h}, \qquad |u_r| = O(\tau^{-\frac{1}{2}}), \qquad \text{volume} \asymp \tau^{\frac{3}{2}-h}$$

Quantity	Scale	Meaning
kinetic energy	$\tau^{1/2-3h} \to 0$	energy stays bounded
dissipation rate	$\tau^{-1/2-3h}$	integrable since $h < 1/6$
swirl Reynolds number	$\tau^{-h} \to \infty$	many turns per diffusion time
radial Reynolds number	$O(1)$	viscosity stays in the balance
$\sqrt{\tau}\,\|u\|_{L^\infty}$	$\tau^{-h} \to \infty$	faster than self-similar

$$R_\theta^{(0)} = -(\partial_r + \tfrac{2}{r})\, T_{r\theta}, \qquad R_z^{(0)} = -(\partial_r + \tfrac{1}{r})\, T_{rz}$$

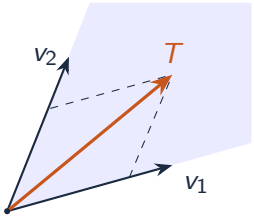
- core and heat exterior balance their own momentum
- the joining annulus $X_a < X < X_b$ leaves a residual that blows up
- it is a **radial flux** of swirl and axial momentum: a stress T
- matching **five radial integrals** keeps T supported in the annulus
- a small oscillation $\sim N^{-1}\sin(N \log X)$ steers the shear so that T lies in the **admissible stress cone**

Pulses supply the stress [Paper §3.3; Heuristic]

$$\mathcal{R}(u_B + w) = \mathcal{R}(u_B) + L_{u_B} w + \nabla \cdot (w \otimes w)$$

Plane wave $w = a\cos(\xi \cdot x + \varphi)$, $a \cdot \xi = 0$:

- $\nabla \cdot w = 0$ and $(w \cdot \nabla) w = 0$
- average stress $\langle w \otimes w \rangle = \frac{1}{2}\, a \otimes a$
- slowly varying amplitude $\Rightarrow$ prescribable force
- squared amplitudes are positive, so two families reach exactly the cone $c_1 v_1 + c_2 v_2$, $c_i > 0$

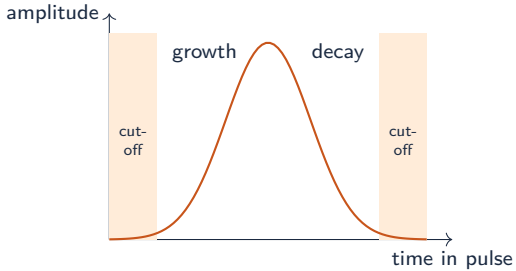


Where pulses get their energy [Heuristic; Paper §2.2]

Rayleigh's toy model for swirl $V(r)$:

$$\ddot{w}_r = -\frac{k_z^2}{|k|^2}\,\kappa^2 w_r, \qquad \kappa^2 = \frac{1}{r^3}\frac{d}{dr}(rV)^2$$

- angular momentum decreasing outward $\Rightarrow \kappa^2 < 0 \Rightarrow$ growth
- shear tilts wave fronts, so $|k_r|$ grows
- amplification weakens, damping $\nu|k|^2$ grows
- each pulse rises, then decays



Correcting to infinite order [Paper §3.4, §9]

Each cycle, after which the full residual is recomputed:

1. solve wave-amplitude equations
2. adjust amplitudes to fix the averaged stress
3. correct the angular mean flow
4. restore the five radial moments

$$\sigma_0 = \tfrac{1}{5}, \qquad \sigma_{j+1} = \sigma_j + \tfrac{1}{10} \;\to\; \infty$$

Summation

Corrections are summed with cutoffs whose supports shrink toward the singular point (Borel-type). The residual becomes **flat**: $O(q^N)$ for every N, with all derivatives.

Finishing the proof [Paper §10]

1. **Localize:** $u = \operatorname{curl}(cA) + cBe_\theta$, so the flow is divergence-free and compactly supported
2. **Extend the force:** all derivatives have limits at $t = 1$, so a Borel lemma extends f smoothly past $t = 1$
3. **Energy:** $\|u(t)\|^2 + 2\int_0^t \|\nabla u\|^2 \le \left(\int_0^t \|f\| \right)^2$
4. **Uniqueness:** any smooth bounded-energy solution equals u before $t = 1$ (pressure via Riesz transforms), so it also blows up
5. **All ν and the torus:** $u_\nu = \sqrt{\nu}\, u(x/\sqrt{\nu}, t)$; shrink, then periodize $U = \sum_{k \in \mathbb{Z}^3} \tilde{u}(x + k, t)$

What the Lean formalization certifies

```
navier_stokes_breakdown_R3 (nu) (hnu :  nu > 0) :
  ∃ u0 f, InitialVelocityConditionDecay u0 ∧ ForceConditionDecay f ∧
    ¬∃ v p, NavierStokesExistenceAndSmoothnessRn nu u0 f v p
```

- statement adapted from Google DeepMind *Formal Conjectures*; periodic analogue for (D)
- proofs in `openai/NavierStokesAndEuler` @ `f9e8bc5b`, Lean v4.34.0-rc2
- Comparator: only `propext`, `Quot.sound`, `Classical.choice`
- certifies: the stated theorems follow from Lean + Mathlib
- a human must still check that the statement matches the Clay text
- this exposition did not rebuild the project

Summary

1. collapse an anisotropic vortex
2. let amplified pulses supply its missing momentum flux
3. correct to infinite order, so the force is smooth through $t = 1$
4. localize, extend the force, prove uniqueness

Open

Unforced alternatives (A) and (B). Independent review of the 166-page proof.

Essay (v1.0): github.com/Thanh-Van-2001/navier-stokes-blowup-exposition
Source: OpenAI, *Finite time blowup for Navier–Stokes* (2026); Lean: github.com/openai/NavierStokesAndEuler